# Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

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**Abstract**

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**Keywords:** economic commensalism; environmental economics; rational incentives; commons; carbon markets.

## 1. Introduction

From an intertemporal social-efficiency perspective, neither extreme predation, defined by the indiscriminate and often illegal depletion of natural resources, nor strict preservationism, characterized by minimal or non-extractive use of the environment, can provide a stable and efficient resolution of the trade-off between production and ecological integrity (DASGUPTA, 2001). These two poles represent opposing boundaries of environmental interaction: the former disregards sustainability entirely, while the latter may overly constrain productive potential under conventional productivity metrics, which typically emphasize output expansion over ecological feedbacks. Between these extremes lies the economic problem of resource extraction: the allocation of renewable and non-renewable

natural resources over time, under conditions of scarcity, regeneration, depletion, opportunity costs, and intertemporal trade-offs (HOTELLING, 1931; CLARK, 2010; DASGUPTA, 2001). In this narrower economic sense, extraction need not be predatory; it may remain compatible with sustainability when the rate of use respects regenerative capacity, stock constraints, and the shadow value of future resource availability.

The concept of extractivism, however, carries a different meaning in the political economy literature. Rather than denoting resource extraction as an intertemporal allocation problem, extractivism refers to a development pattern based on the large-scale appropriation of natural resources, frequently oriented toward commodity exports, limited domestic value addition, and ecological degradation (GUDYNAS et al., 2009; ACOSTA, 2013; MCKAY, 2017). Under this interpretation, the issue is not extraction per se, but the institutional and productive regime through which extraction becomes structurally tied to depletion, dependency, and asymmetric distribution of environmental costs.

–> Mostrar que a economia tradicional não ignora completamente a natureza, mas tende a tratá-la como estoque, restrição, externalidade, renda da terra ou custo ambiental;

–> Dizer que o problema do artigo é outro: a natureza raramente aparece como fator produtivo endógeno, dinâmico e estrategicamente incorporado à decisão econômica;

–> Introduzir, com cautela, a lacuna que justificará o economic commensalism. This distinction is central to the argument developed here. The paper does not treat extraction and extractivism as interchangeable terms. Instead, it uses the gap between them to identify the conditions under which resource use moves from an economically rational allocation problem to an ecologically predatory regime, as reflected in declining future productivity, ecological threshold violations, divergence between private and social payoffs, and intertemporal welfare losses.

## 2. The Limits of Current Environmental Governance

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

## 3. Literature Review: From Externalities to Incentives

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## 4. Model

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## 5. Results

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## 6. Discussion

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## 7. Final Remarks

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